§1.571 — AC regular via idempotent splitting

Reading of Freyd & Scedrov, *Categories and Allegories* §1.571, with the construction drawn out. Composition in **diagram order**: `xy` means first `x` then `y`. Companion to the Lean proof `ac_factorization_via_idempotent` in `Fredy/S1_57.lean`, and to the pullback/equalizer background in `Fredy/S1_57.md`.

The claim

§1.571 gives a sufficient condition for a Cartesian category to be **AC regular** — using the alternative definition from §1.57: *every morphism factors as a left-invertible followed by a monic.*

Hypothesis. For every $x : A \to B$ there is an idempotent $e : A \to A$ with

- $e^2 = e$,
- $ex = x$ (`e` fixes `x`: first `e`, then `x`), and
- ***e* and *x* have the same level.**

What "level" means

The *level* of x is its **kernel pair**: the pullback of x against itself. Its elements are the pairs (a, a') that x merges.

$$\begin{tikzcd}
\ker(x) \arrow[r, "k_2"] \arrow[d, "k_1"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "x"] \\
A \arrow[r, "x"] & B
\end{tikzcd}$$

$\ker(x) = \{(a, a') : xa = xa'\} \subseteq A \times A$ — *the pairs x identifies.*

"e and x have the **same level**" means $\ker(e) = \ker(x)$ as subobjects of $A \times A$: e merges a pair iff x does. This is an **assumption** (Lean: both inclusions `kernelPairRel e ⊂ kernelPairRel x` and back); the construction below *uses* it.

Decoding the paragraph

> Let $C \to A$ be an equalizer of $1, e$. Let $(A \to C \to A) = e$. Then $x = (A \to C \to A \to B)$. $A \to C$ has a left-inverse $(C \to A)$ and $C \to A \to B$ is monic.

Let $m : C \to A$ be the equalizer of 1_A and e.

Step 1–3: split the idempotent through the equalizer

Since $e; e = e = e; 1$, the map e *equalizes* the parallel pair $1, e$, so it factors **uniquely** through the equalizer m: there is a unique $p : A \to C$ with $p; m = e$ (unique because m is monic).

$$\begin{tikzcd}
C \arrow[r, tail, "m"] & A \arrow[r, shift left, "1"] \arrow[r, shift right, "e"'] & A \\
& A \arrow[u, "e"'] \arrow[ul, dashed, "p"] &
\end{tikzcd}$$

The apex A is a cone over $1, e$ (since $e; 1 = e = e; e$); it factors uniquely through the equalizer m by the dashed p, with $p; m = e$.

This **splits the idempotent** $e = p; m$. Moreover $m; p = 1_C$ — because $m; p; m = m; e = m = 1; m$ and m is monic. So ***m* is a left-inverse of *p*** ("$A \to C$ has a left-inverse $C \to A$"), making p left-invertible (a cover). Equalizer maps are always monic (`eqMap_mono`), so m is monic.

Left-invertible vs cover vs epic

These three "surjective-like" notions are distinct; left-invertible is the strongest, and it is the one §1.57 demands (which is the choice content).

Notion	Meaning (diagram order)	Splits?	This, but not the next-stronger
Epic	$pu = pv \Longrightarrow u = v$	—	$\mathbb{Z} \to \mathbb{Q}$ in **Ring**: epic, yet not onto (not a cover)
Cover (regular epi)	coequalizer of its kernel pair; pullback-stable	no	$\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ in **Grp**: onto, but no section
Left-invertible (split epi)	has a section m with $m; p = 1$	yes	$A \times B \twoheadrightarrow A$ (B nonempty): section $a \mapsto (a, b_0)$

*Bottom $\Longrightarrow$ top: split epi $\Longrightarrow$ cover $\Longrightarrow$ epic; each implication is strict (the examples show why). In **Set** with choice all three coincide — that is exactly why Set is AC regular.*

Step 4–5: factor x, and show the second leg is monic

From $ex = x$ and $e = p; m$ we get $x = ex = p; (m; x)$. Set $n := m; x$. Then $x = p; n$ — the required left-invertible-then-monic factorization:

$$\begin{tikzcd}
A \arrow[r, two heads, "p"] \arrow[rr, bend right, "x"'] & C \arrow[r, tail, "n = mx"] & B
\end{tikzcd}$$

$x = p; n$ with p left-invertible (a cover) and $n = mx$ monic.

Why $n = mx$ is monic — the **only** step using "same level". First, C is the **fixed points** of e: as the equalizer of $1, e$ it satisfies $m; 1 = m; e$, i.e. $\boldsymbol{me = m}$.

$$\begin{tikzcd}
C \arrow[r, tail, "m"] & A \arrow[r, shift left, "1"] \arrow[r, shift right, "e"'] & A
\end{tikzcd}$$

$m; 1 = m; e$, so $me = m$: e fixes everything in C.

Now test n against any pair $u, v : W \rightrightarrows C$. Since $\ker(x) = \ker(e)$, "x merges" = "e merges", and $me = m$ then strips the e back off:

$$\begin{aligned}
un = vn &\Longrightarrow umx = vmx && (\text{def. } n = mx) \\
&\Longrightarrow ume = vme && (\ker(x) = \ker(e)) \\
&\Longrightarrow um = vm && (me = m) \\
&\Longrightarrow u = v && (m \text{ monic})
\end{aligned}$$

With both legs in hand — p left-invertible, n monic — the category is AC regular.